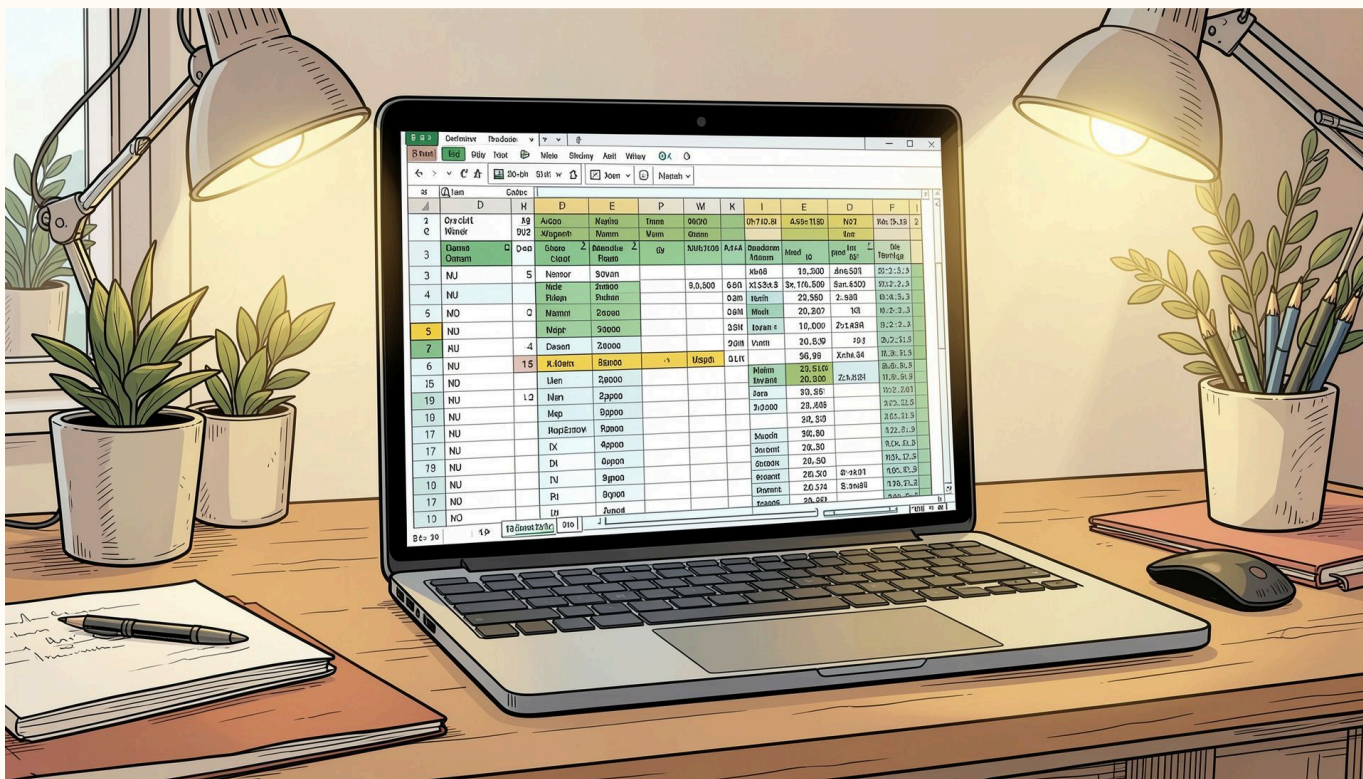


# Customer Shopping Behavior Analysis

Uncovering spending patterns, segments, and preferences from **3,900 transactions** to guide strategic decisions.



# Dataset at a Glance



**3,900 Rows**

Transactions analyzed

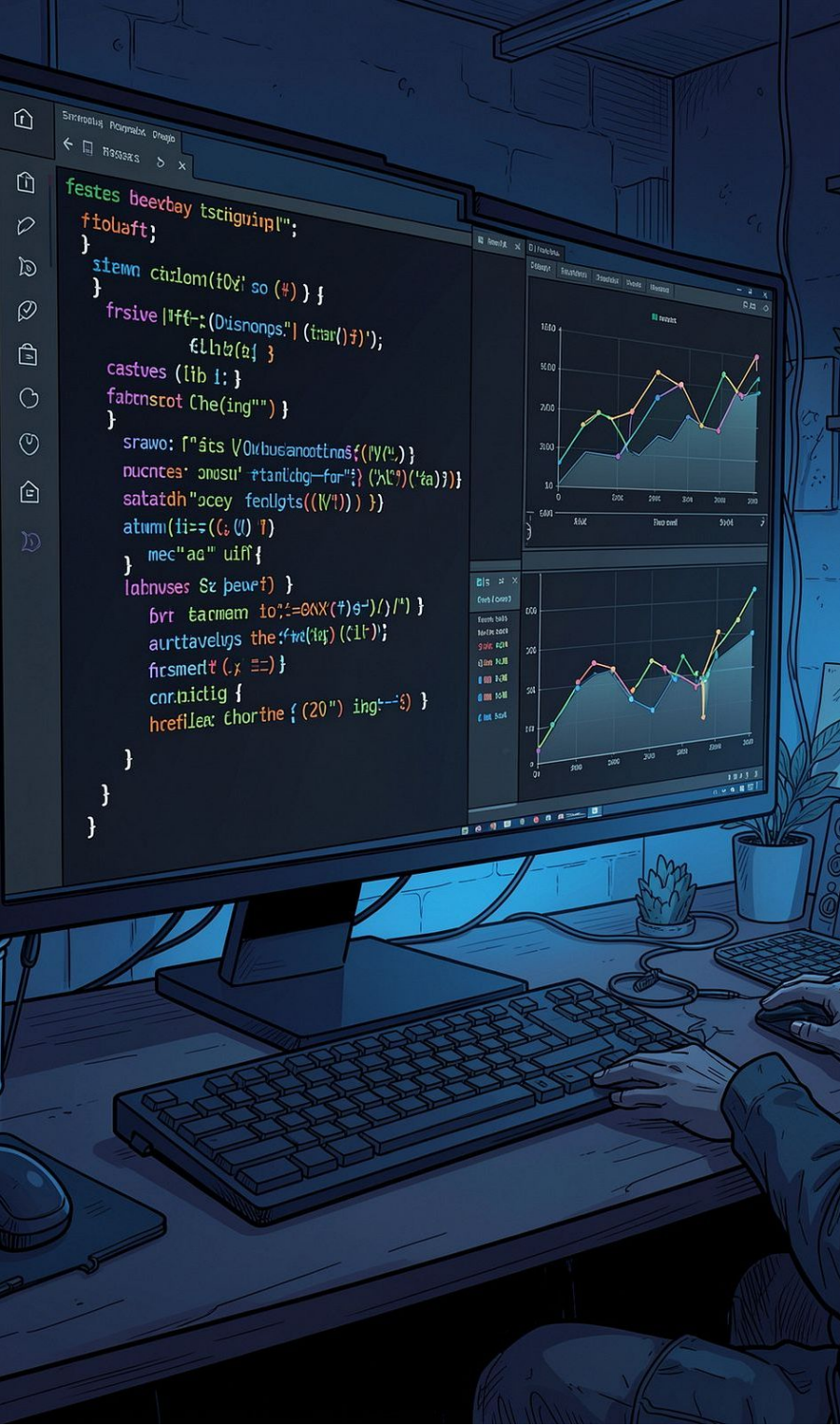
**18 Columns**

Features tracked

**37 Missing**

Review Rating values

Key features: demographics, purchase details, shopping behavior, subscription status.



# Data Preparation in Python

01

## Load & Explore

Imported via `pandas`; `df.info()`, `describe()`

02

## Clean & Standardize

Imputed missing ratings by category median; renamed to `snake_case`

03

## Feature Engineering

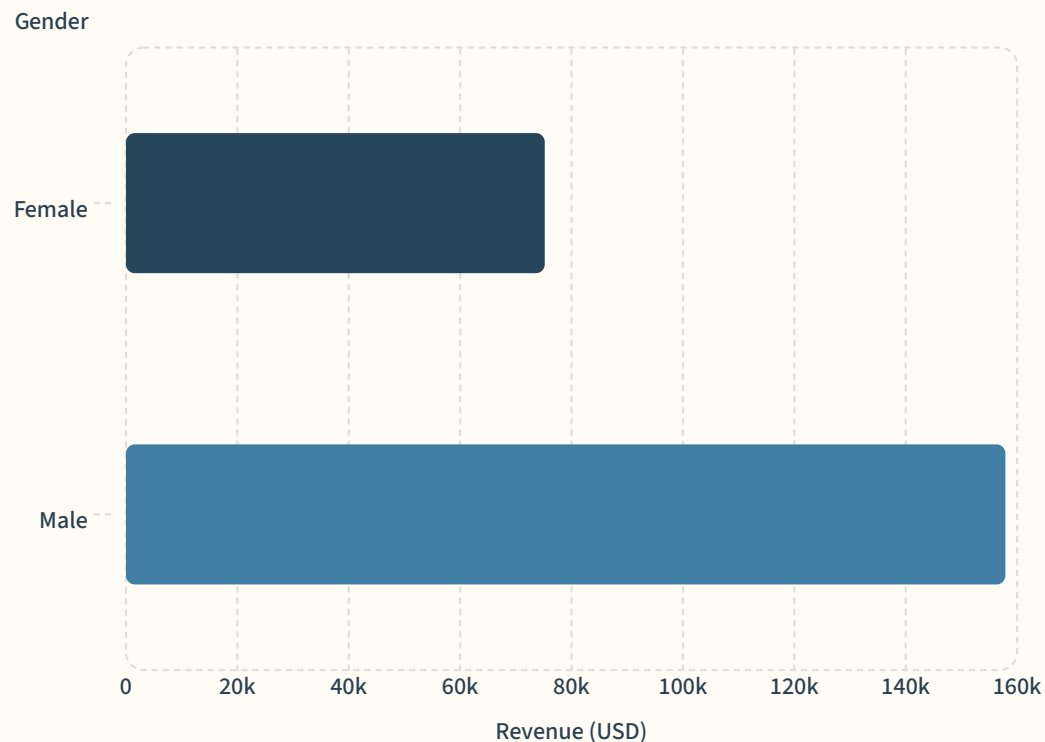
Created `age_group` and `purchase_frequency_days`; dropped redundant `promo_code_used`

04

## Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis

# Revenue by Gender

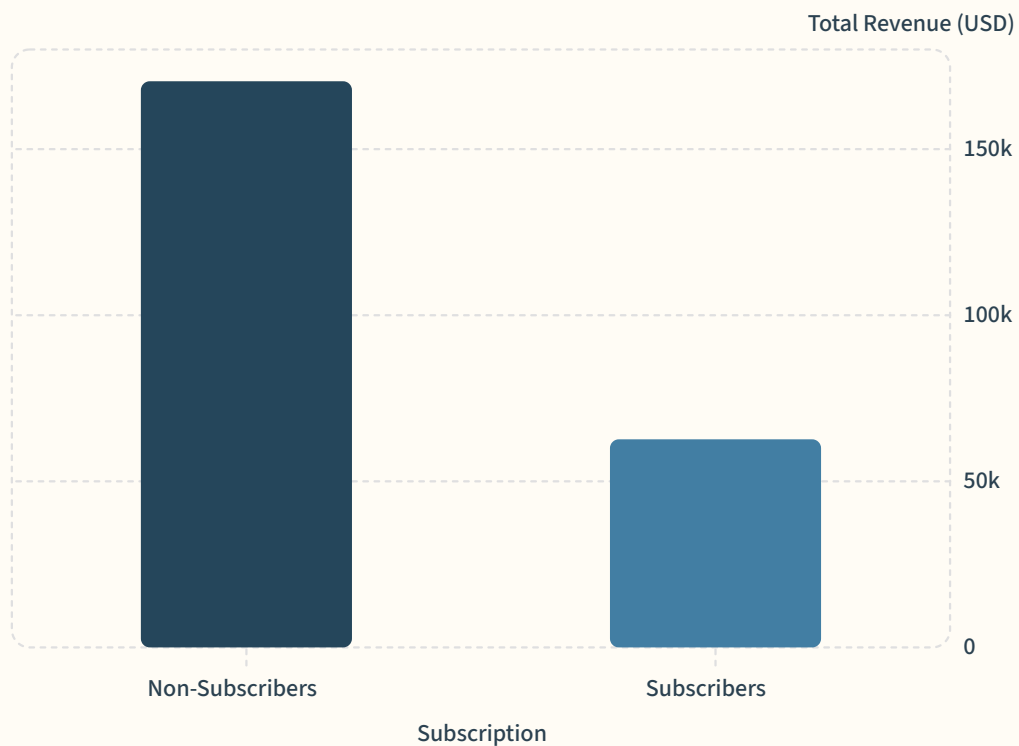


## Key Finding

Male customers generated **2.1× more revenue** than female customers — \$157,890 vs. \$75,191.

- ❏ Male shoppers represent the higher-value segment by total spend.

# Subscribers vs. Non-Subscribers



## Non-Subscribers

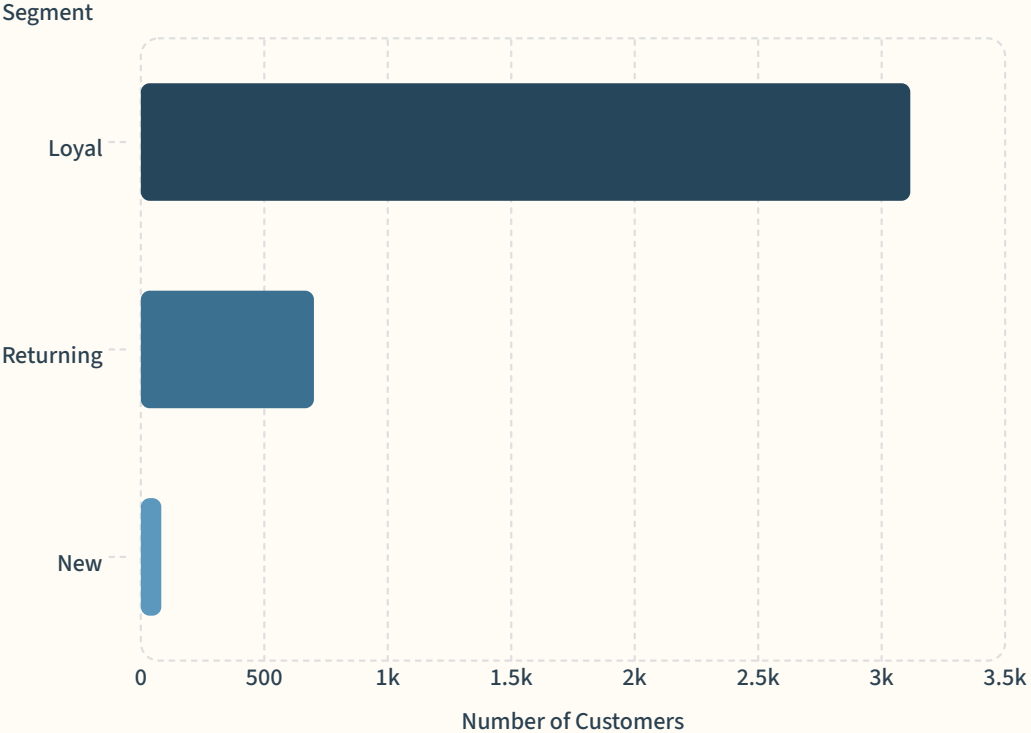
2,847 customers · \$59.87  
avg spend · **\$170,436** total

## Subscribers

1,053 customers · \$59.49  
avg spend · **\$62,645** total

Average spend is nearly identical — subscription drives volume, not basket size.

# Customer Segmentation



## Segment Breakdown

Of 958 repeat buyers (>5 purchases), only 1,053 are subscribers — opportunity to convert.

### Loyal

3,116 customers

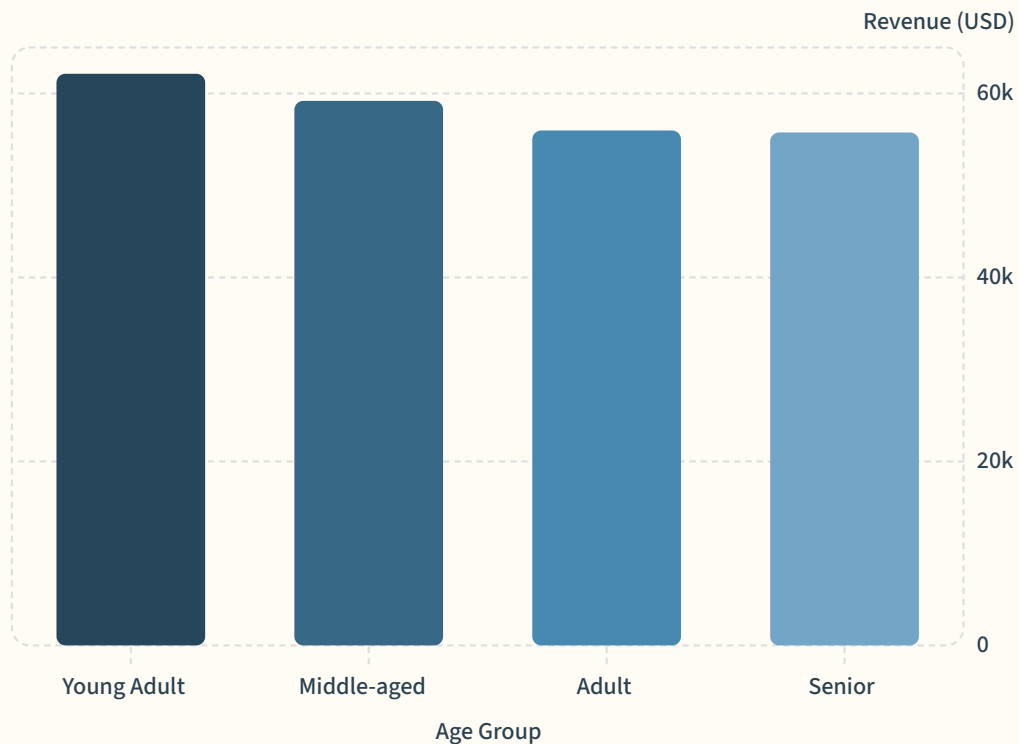
### Returning

701 customers

### New

83 customers

# Revenue by Age Group



## Insight

Young Adults lead revenue at **\$62,143**. Revenue declines gradually with age — all groups remain meaningful contributors.

- ❏ Focus marketing on Young Adult and Middle-aged segments for maximum ROI.

# Top Products & Discount Behavior

Top 5 by Rating

| Item    | Avg Rating |
|---------|------------|
| Gloves  | 3.86       |
| Sandals | 3.84       |
| Boots   | 3.82       |
| Hat     | 3.80       |
| Skirt   | 3.78       |

Most Discounted Items

| Item     | Discount Rate |
|----------|---------------|
| Hat      | 50.00%        |
| Sneakers | 49.66%        |
| Coat     | 49.07%        |
| Sweater  | 48.17%        |
| Pants    | 47.37%        |



# Power BI Dashboard

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

27%

Subscribers

Interactive filters: Subscription Status, Gender, Category, Shipping Type. Revenue leaders: Clothing (~\$90K), Accessories (~\$70K).

# Business Recommendations



## Boost Subscriptions

Promote exclusive subscriber benefits



## Loyalty Programs

Reward repeat buyers to grow Loyal segment



## Review Discount Policy

Balance sales lift with margin control



## Targeted Marketing

Focus on high-revenue age groups & express-shipping users



## Product Positioning

Highlight top-rated & best-selling products